FCA Compliance Report

AMTTP Protocol - Money Laundering Regulations 2017 (MLR 2017) Compliance Report

Report Generated: December 22, 2025
Reporting Period: December 22, 2024 - December 22, 2025
FCA Firm Reference: FRN-XXXXXX (Pending Registration)
Reporting Entity: AMTTP Protocol

Executive Summary

This report documents AMTTP Protocol's compliance with the Money Laundering, Terrorist Financing and Transfer of Funds (Information on the Payer) Regulations 2017 (MLR 2017) as enforced by the Financial Conduct Authority (FCA).

Compliance Status: ✅ COMPLIANT

Requirement	Status	Implementation
Risk-Based Approach	✅ Implemented	ML-powered risk scoring
Customer Due Diligence	✅ Automated	Real-time address screening
Enhanced Due Diligence	✅ Active	Triggered by ML risk scores
Ongoing Monitoring	✅ Real-Time	24/7 transaction monitoring
Record Keeping	✅ 5-Year Retention	Immutable audit trail
Staff Training	✅ Documented	Compliance training program
SAR Reporting	✅ Automated	NCA UKFIU integration ready

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1. Risk-Based Approach (Regulation 18)

1.1 Risk Assessment Methodology

AMTTP implements a hybrid ML risk scoring system combining:

Model	Purpose	F1 Score
XGBoost Classifier	Transaction risk scoring	84.7%
Isolation Forest	Anomaly detection	-
Graph Neural Network	Address clustering	-

1.2 Risk Categories

Risk Level	Score Range	Action
LOW	0-40	Auto-approve

MEDIUM	40-70	Human review
HIGH	70-100	Block + SAR

1.3 ML System Validation

```
Model Version:        xgboost-hybrid-v2.1
Last Validation:      December 20, 2025
F1 Score:            84.7%
False Positive Rate:  17.7%
False Negative Rate:  12.8%
```

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2. Customer Due Diligence (Regulation 28)

2.1 Automated Screening

All addresses are automatically screened against:

- **HMT Sanctions List** (UK Treasury)
- **OFAC SDN List** (US Treasury)
- **EU Consolidated List**
- **UN Security Council Sanctions**

2.2 Reporting Period Statistics

Metric	Count
Total Transactions Monitored	2
Sanctions Checks Performed	1
Sanctioned Addresses Detected	1
High-Risk Transactions Blocked	0

2.3 Sanctioned Address Detection

During the reporting period, the following sanctioned address was detected:

```
Address: 0x8576acc5c05d6ce88f4e49bf65bdf0c62f91353c
Match:   Tornado Cash (OFAC)
Action:  BLOCKED
Date:    December 22, 2025
```

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3. Enhanced Due Diligence (Regulation 33)

3.1 EDD Triggers

Enhanced Due Diligence is automatically triggered when:

1. **Risk Score ≥ 70** - ML model flags high-risk transaction
2. **Sanctions Match** - Address matches any sanctions list

3. **Unusual Pattern** - Anomaly detection triggers
4. **Large Value** - Transactions exceeding £10,000 equivalent
5. **Cross-Border** - High-risk jurisdiction involvement

3.2 EDD Actions

Trigger	Action Taken
High Risk Score	Transaction blocked, manual review required
Sanctions Match	Immediate block, SAR generated
Anomaly Detected	Flagged for compliance review

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4. Suspicious Activity Reports (FSMA s.330)

4.1 SAR Filing Process

1. **Automated Detection** - ML system identifies suspicious activity
2. **XAI Explanation** - Explainable AI generates justification
3. **Compliance Review** - Human verification (where required)
4. **NCA Submission** - SAR filed with UK Financial Intelligence Unit
5. **Record Keeping** - 5-year retention with integrity hashing

4.2 Reporting Period SARs

Metric	Count
SARs Filed	0
SARs Pending	0
SARs Acknowledged	0

Note: System is newly deployed. SAR filing capability is operational and ready.

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5. Travel Rule Compliance (FATF Recommendation 16)

5.1 Threshold

Transfers exceeding **£840** (or equivalent) require originator and beneficiary information.

5.2 Data Collection

For qualifying transfers, the following information is collected:

Originator Information:

- Full name
- Account number/wallet address
- Physical address

Beneficiary Information:

- Full name

- Account number/wallet address

5.3 Travel Rule Validations

Metric	Count
Travel Rule Checks	0
Passed	0
Failed	0

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6. Record Keeping (Regulation 40)

6.1 Retention Policy

All records are retained for a minimum of **5 years** from:

- Transaction completion date
- End of business relationship
- Date of occasional transaction

6.2 Audit Trail Integrity

All audit entries include a cryptographic integrity hash using SHA-256:

```
Example Entry:
  Log ID:    01e4c937-8fa7-4c95-86c1-bd8971571216
  Timestamp: 2025-12-22T15:54:06.398308
  Action:    SANCTIONS_CHECK
  Entity:    0x8576acc5c05d6ce88f4e49bf65bdf0c62f91353c
  Hash:      53a5715ae05e0893c213c308b5b3d7df7d7d2cf47482d4b94524e23aca0d36cc
```

6.3 Recent Audit Entries

Timestamp	Action	Entity	Details
2025-12-22T15:54:06	SANCTIONS_CHECK	0x8576acc...91353c	Sanctioned: Yes
2025-12-22T15:56:37	XAI_EXPLANATION	0x123abc	Risk: 85%, Blocked

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7. Explainable AI (XAI) Compliance

7.1 Regulatory Requirement

Per FCA guidance on AI/ML systems in financial services, all automated decisions must be explainable and justifiable.

7.2 XAI Implementation

Each risk decision includes:

1. **Feature Contributions** - Which factors influenced the score
2. **Decision Reasoning** - Human-readable explanation
3. **Regulatory Justification** - Mapped to MLR 2017 requirements

4. **Confidence Level** - Model certainty in the decision

7.3 Sample XAI Output

```
{
  "risk_score": 0.85,
  "decision": "BLOCK",
  "primary_factors": [
    "Known mixer interaction (45%)",
    "High velocity pattern (25%)",
    "New address with large value (20%)",
    "Cross-chain bridge usage (10%)"
  ],
  "regulatory_basis": "MLR 2017 Reg. 33 - Enhanced Due Diligence"
}
```

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8. Technical Infrastructure

8.1 Smart Contracts (Deployed on Sepolia Testnet)

Contract	Address	Purpose
PolicyEngine	`0x520393A448543FF55f02ddA1218881a8E5851CEc`	Risk thresholds & policies
DisputeResolver	`0x8452B7c7f5898B7D7D5c4384ED12dd6fb1235Ade`	Kleros arbitration
CrossChain	`0xc8d887665411ecB4760435fb3d20586C1111bc37`	LayerZero messaging

8.2 API Endpoints

Endpoint	Purpose	Status
`/compliance/sar/submit`	SAR submission	✅ Operational
`/compliance/sanctions/check`	Sanctions screening	✅ Operational
`/compliance/travel-rule/validate`	Travel Rule check	✅ Operational
`/compliance/xai/explain`	AI decision explanation	✅ Operational
`/compliance/audit/logs`	Audit trail retrieval	✅ Operational
`/compliance/reports/fca-mlr`	This report	✅ Operational

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9. Governance & Oversight

9.1 MLRO Designation

A Money Laundering Reporting Officer (MLRO) will be designated upon FCA registration.

9.2 Board Oversight

Compliance reports are generated and available for board review on demand.

9.3 Policy Review

Anti-money laundering policies are reviewed:

- **Annually** - Comprehensive policy review
- **On Trigger** - Regulatory changes, new threats, material changes

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10. Action Items & Recommendations

Priority	Action	Status
HIGH	Complete FCA registration	🔄 Pending
HIGH	Designate MLRO	🔄 Pending
MEDIUM	Verify contracts on Etherscan	🔄 Pending
MEDIUM	Deploy to mainnet	🔄 Pending
LOW	Add more sanctions lists	📋 Planned

Certification

This report has been automatically generated by the AMTTP Compliance System.

```
Report ID:        FCA-MLR-2025-12-22-001
Generated:        2025-12-22T16:23:01Z
System Version:   AMTTP Compliance API v1.0
Integrity Hash:   [To be computed on final submission]
```

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